Homework 2: Supply and Economy with Production

Sergey V. Popov

2026-08-10

Due: Sep 7, 2026, noon.

Production set

Consider a space of bundles $(x_1, x_2) \in R^2$. Let one firm have access to technology that makes at most $2\sqrt{-y_2}$ units of good 1 when $-y_2$ units of good 2 is spent (notice that y_2 is negative), but it can't make y_2 out of y_1. Another firm has a technology that makes at most $3\sqrt{-y_1}$ units of the second good if $-y_1$ units of good 1 are spent.

- Plot production sets of these firms (Y_1 for the first firm, Y_2 for the second). Are they convex? Do they feature decreasing returns to scale?
- Solve for each firm's profit function.
- Imagine now these two firms merged into one firm. Explain how would the production set of the merged firm look like.

Stuff and Leisure

The economy contains

- a representative consumer with preferences over Stuff and Leisure with an endowment of 20 units of Leisure (that can be sold to the firm as Labour)
- and a firm owned by said consumer that produces $5\sqrt{L}$ of Stuff when L units of Labour are used.

Calculate

- the cost function of the firm and its production point given wage and price; observe that the production point is a function of the real wage w/p;
- the Walrasian equilibrium price and quantity if the utility of the consumer over Stuff and Leisure is $U(S, L) = 10S - S^2 + L$.

What happens to the real wage if the leabour supply increases? Why?